

Surface-Mount Schottky Barrier Rectifier


SMA (DO-214AC)

Cathode  Anode

LINKS TO ADDITIONAL RESOURCES


[3D Models](#)

PRIMARY CHARACTERISTICS

$I_{F(AV)}$	1.5 A
V_{RRM}	25 V, 35 V, 45 V
I_{FSM}	40 A
V_F	0.50 V
T_J max.	150 °C
Package	SMA (DO-214AC)
Circuit configuration	Single

FEATURES

- Low profile package
- Ideal for automated placement
- Guardring for overvoltage protection
- Low power losses, high efficiency
- Very low switching losses
- High surge capability
- Meets MSL level 1, per J-STD-020, LF maximum peak of 260 °C
- AEC-Q101 qualified available
 - Automotive ordering code: base P/NHE3 or P/NHM3
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912


RoHS
COMPLIANT
HALOGEN
FREE
Available

TYPICAL APPLICATIONS

For use in low voltage high frequency inverters, freewheeling, DC/DC converters, and polarity protection applications.

MECHANICAL DATA

Case: SMA (DO-214AC)

Molding compound meets UL 94 V-0 flammability rating

Base P/N-E3 - RoHS-compliant, commercial grade

Base P/N-M3 - halogen-free, RoHS-compliant, commercial grade

Base P/NHE3_X - RoHS-compliant and AEC-Q101 qualified
Base P/NHM3_X - halogen-free, RoHS-compliant, and AEC-Q101 qualified

("X" denotes revision code e.g. A, B,

Terminals: matte tin plated leads, solderable per J-STD-002 and JESD 22-B102

E3, M3, HE3, and HM3 suffix meets JESD 201 class 2 whisker test

Polarity: color band denotes the cathode end

MAXIMUM RATINGS ($T_A = 25\text{ °C}$ unless otherwise noted)

PARAMETER	SYMBOL	BYS10-25	BYS10-35	BYS10-45	UNIT
Device marking code		BYS 025	BYS 035	BYS 045	
Maximum repetitive peak reverse voltage	V _{RRM}	25	35	45	V
Maximum average forward rectified current	I _{F(AV)}	1.5			A
Peak forward surge current single half sine-wave superimposed on rated load	I _{FSM}	40			A
		30			
Junction and storage temperature range	T _J , T _{STG}	-65 to +150			°C

**ELECTRICAL CHARACTERISTICS** ($T_A = 25\text{ °C}$ unless otherwise noted)

PARAMETER	TEST CONDITIONS	SYMBOL	BYS10-25	BYS10-35	BYS10-45	UNIT
Maximum instantaneous forward voltage ⁽¹⁾	1.0 A	V_F	500			mV
Maximum DC reverse current ⁽¹⁾	V_{RRM}	$T_J = 25\text{ °C}$	500			μA
		$T_J = 100\text{ °C}$	10			mA

Note

⁽¹⁾ Pulse test: 300 μs pulse width, 1 % duty cycle

THERMAL CHARACTERISTICS ($T_A = 25\text{ °C}$ unless otherwise noted)

PARAMETER	SYMBOL	BYS10-25	BYS10-35	BYS10-45	UNIT
Maximum thermal resistance, junction-to-lead	R _{θJL}	25			°C/W
Maximum thermal resistance, junction-to-ambient	R _{θJA} ⁽¹⁾	150			°C/W
	R _{θJA} ⁽²⁾	125			
	R _{θJA} ⁽³⁾	100			

Notes

⁽¹⁾ Mounted on epoxy-glass hard tissue

⁽²⁾ Mounted on epoxy-glass hard tissue, 50 mm² 35 μm Cu

⁽³⁾ Mounted on Al-oxide-ceramic (Al₂O₃), 50 mm² 35 μm Cu

ORDERING INFORMATION (Example)

PREFERRED P/N	UNIT WEIGHT (g)	PREFERRED PACKAGE CODE	BASE QUANTITY	DELIVERY MODE
BYS10-45-E3/TR	0.064	TR	1800	7" diameter plastic tape and reel
BYS10-45-E3/TR3	0.064	TR3	7500	13" diameter plastic tape and reel
BYS10-45HE3_A/H ⁽¹⁾	0.064	H	1800	7" diameter plastic tape and reel
BYS10-45HE3_A/I ⁽¹⁾	0.064	I	7500	13" diameter plastic tape and reel
BYS10-45-M3/TR	0.064	TR	1800	7" diameter plastic tape and reel
BYS10-45-M3/TR3	0.064	TR3	7500	13" diameter plastic tape and reel
BYS10-45HM3_A/H ⁽¹⁾	0.064	H	1800	7" diameter plastic tape and reel
BYS10-45HM3_A/I ⁽¹⁾	0.064	I	7500	13" diameter plastic tape and reel

Note

⁽¹⁾AEC-Q101 qualified



RATINGS AND CHARACTERISTICS CURVES ($T_A = 25^\circ\text{C}$ unless otherwise noted)

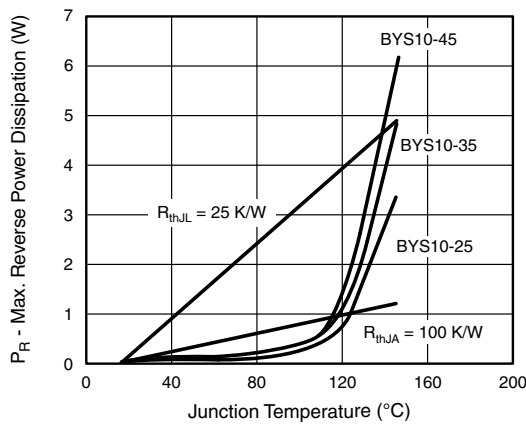


Fig. 1 - Max. Reverse Power Dissipation vs. Junction Temperature

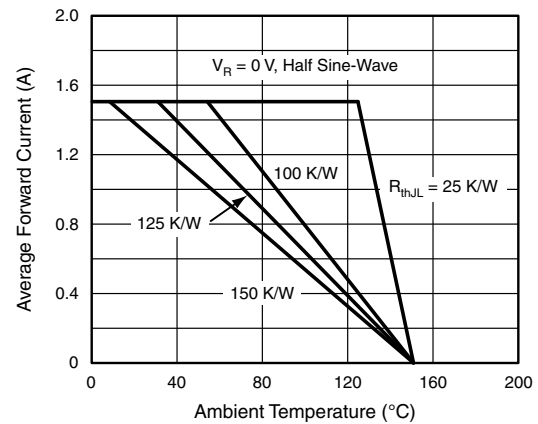


Fig. 4 - Max. Average Forward Current vs. Ambient Temperature

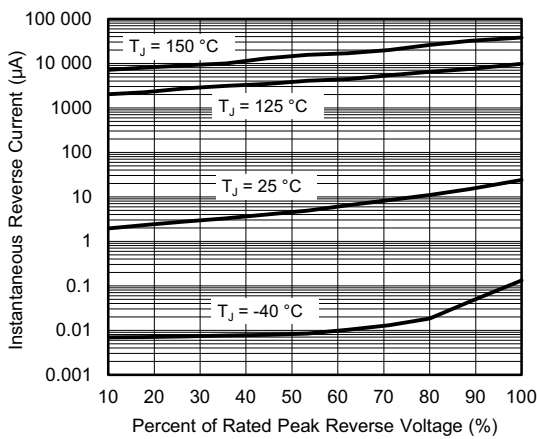


Fig. 2 - Typical Reverse Characteristics

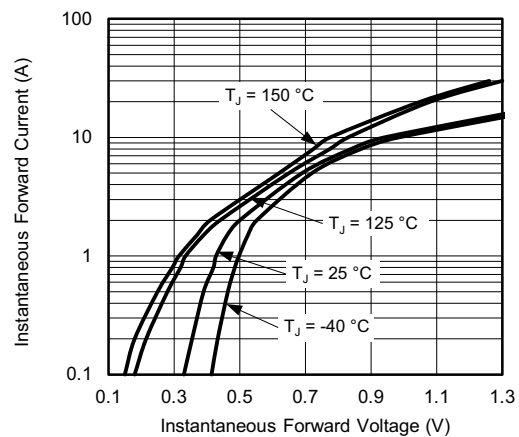


Fig. 5 - Typical Instantaneous Forward Characteristics

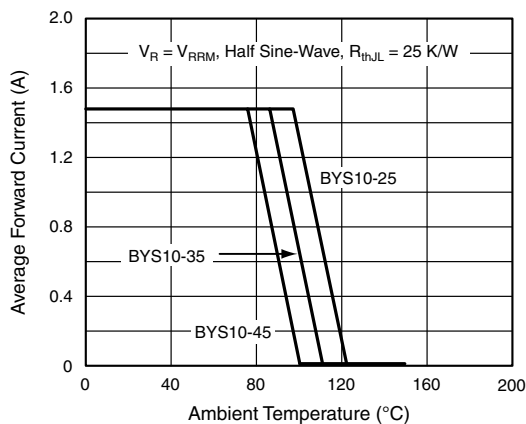


Fig. 3 - Max. Average Forward Current vs. Ambient Temperature

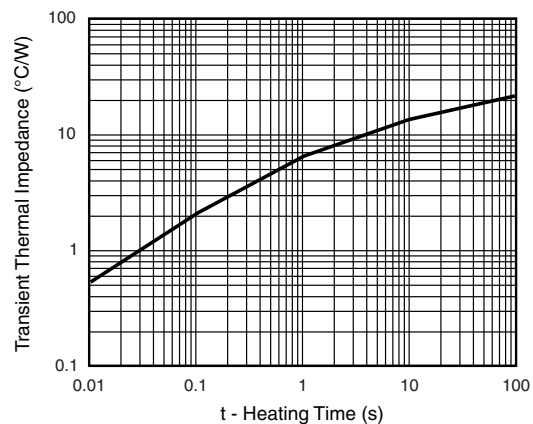
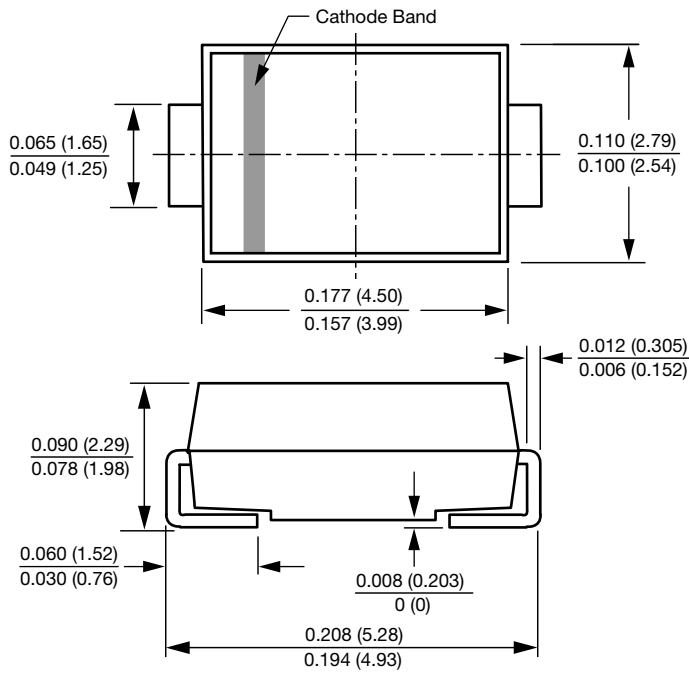


Fig. 6 - Typical Transient Thermal Impedance

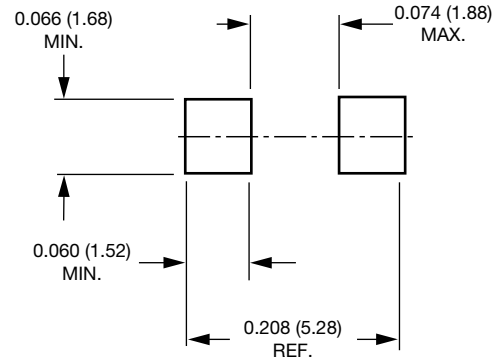


PACKAGE OUTLINE DIMENSIONS in inches (millimeters)

SMA (DO-214AC)



Mounting Pad Layout





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